Snippet 1

- **Time Complexity:** O(n)
- **Space Complexity:** O(1)
- **Justify:** The loop checks each element of the array once, taking linear time, while only a few variables (max_val, i) are used, requiring constant extra space.

Snippet 2

- **Time Complexity:** $O(n^2)$
- **Space Complexity:** O(1)
- **Justify:** The two nested loops compare every possible pair of elements, resulting in quadratic time, while only loop variables are used, requiring constant extra space.

Snippet 12

- **Time Complexity:** O(n)
- **Space Complexity:** O(n)
- **Justify:** The function makes one recursive call for each value from n down to 1, giving linear time, and the recursion stack can contain up to n calls, requiring linear space.

Snippet 13

- **Time Complexity:** O(2^n)
- **Space Complexity:** O(n)
- **Justify:** Each call recursively creates two more calls (n-1 and n-2), producing exponential time, while the maximum recursion depth is n, requiring linear stack space.